Mathematical Modeling of Epidemic Spread Using the SIR Model

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Abstract

This project investigates the dynamics of infectious disease transmission by applying Forward Euler and fourth-order Runge-Kutta (RK4) numerical solvers to empirical datasets, demonstrating that while the standard SIR model accurately forecasts single-wave outbreaks, its structural constraints inherently fail to capture complex, multi-wave pandemics.

1 Introduction

The mathematical modeling of infectious diseases is a critical application of scientific computing, providing the quantitative frameworks necessary to understand and simulate public health crises. This project explores the dynamics of epidemic spread through the Susceptible-Infectious-Recovered (SIR) compartmental model, one of the most fundamental frameworks in mathematical epidemiology.

Because the dynamic transitions between population compartments are governed by a system of non-linear ordinary differential equations (ODEs) that lack simple analytical solutions, numerical approximation is required. To achieve this, the project implements and compares foundational numerical methods, specifically the Forward Euler method and the fourth-order Runge-Kutta (RK4) algorithm. These solvers are rigorously evaluated for their accuracy, numerical stability, and error convergence across different time steps.

To ground the mathematical theory in empirical reality, the computational models are tested against two distinct historical datasets. First, the numerical solvers are validated using a classic benchmark dataset: the 1978 influenza outbreak within a closed British boarding school. This perfectly isolated environment provides an ideal baseline for evaluating the deterministic accuracy of the ODE solvers on a single epidemic wave.

Finally, the project critically examines the inherent limitations of the basic SIR framework when applied to complex, long-term epidemiological events. By evaluating the model against the 2021 COVID-19 multi-wave outbreak in Serbia—utilizing daily incidence data sourced from Our World in Data (OWID)—the study highlights the structural constraints of a closed system. Through this analysis, the project demonstrates both the power of numerical methods in successfully simulating localized outbreaks and their mathematical optimization breakdown when faced with dynamic human interventions and evolving viral transmission rates over prolonged periods.

2 The SIR Model

The Susceptible-Infectious-Recovered (SIR) model stands as one of the most fundamental and influential frameworks in mathematical epidemiology, serving as the cornerstone for understanding infectious disease dynamics across diverse populations and contexts.

The model is based on dividing the host population (humans, in the case of this project) into a small number of compartments, each containing individuals that are identical in terms of their status with respect to the disease in question. In the SIR model, there are three primary compartments:

- Susceptible (S): Individuals who have no immunity to the infectious agent, so they might become infected if exposed.
- Infectious (I): Individuals who are currently infected and can transmit the infection to susceptible individuals whom they contact.
- Removed (R): Individuals who are immune to the infection (either through recovery or death), and consequently do not affect the transmission dynamics in any way when they contact other individuals.

It is traditional to denote the number of individuals in each of these compartments as S, I, and R, respectively. The total host population size is defined as $N = S + I + R$.

Having compartmentalized the population, the next step is to define the mathematical rules that govern how individuals transition between these states over time. These transitions are expressed as a system of non-linear ordinary differential equations (ODEs) that capture the specific rates of infection and recovery, allowing us to simulate the continuous spread and eventual decline of the disease.

2.1 Basic Assumptions

The classic SIR model relies on several key simplifying assumptions:

- **Constant Population:** The total population N remains constant over time ($N = S + I + R$). Births, migrations, and deaths by other causes are ignored.
- **Homogeneous Mixing:** It is assumed that every individual has an equal probability of coming into contact with any other individual in the population.
- **Permanent Immunity:** Once an individual recovers, they acquire lifelong immunity and cannot be reinfected.

2.2 Model Parameters

The flow of individuals between the compartments is governed by two primary rates:

- β (Transmission Rate): This is the 'per capita' rate at which the infection is transmitted from an infectious individual to a susceptible one. In broader epidemiological terms, β is usually understood as the product of two distinct factors: the average number of contacts a person makes per unit of time, and the probability that a disease is actually transmitted during one of those contacts.

- γ (Recovery Rate): This parameter represents the rate at which infected individuals recover (or are otherwise removed from the infectious pool, such as through isolation or death). Its reciprocal, $1/\gamma$, represents the mean infectious period—the average amount of time an individual remains capable of transmitting the disease before moving to the removed compartment.

- N (Total Population Size): The total number of individuals in the host population, defined as $N = S + I + R$. As your text notes, for the differential equations to treat S, I, and R as smooth continuous variables rather than discrete integers, N must be sufficiently large.

- R_0 (Basic Reproduction Number): Defined in your text as $R_0 = \frac{\beta N}{\gamma}$, this is the most critical threshold quantity in epidemic modeling. It represents the expected number of secondary infections generated by a single infected individual introduced into a completely susceptible population ($S(0) = N$).
  - If $R_0 > 1$, each infection spawns more than one new infection, meaning the disease will spread and an epidemic will occur.
  - If $R_0 < 1$, the number of cases will decrease over time, and the outbreak will naturally die out.

2.3 System of Differential Equations

The dynamic transitions between the Susceptible (S), Infectious (I), and Recovered (R) states are mathematically described by the following system of non-linear ordinary differential equations:

$$\frac{dS}{dt} = -\frac{\beta SI}{N} \tag{1}$$

$$\frac{dI}{dt} = \frac{\beta SI}{N} - \gamma I \tag{2}$$

$$\frac{dR}{dt} = \gamma I \tag{3}$$

Let us now explain the equations presented above.

The first equation (1) represents the rate of change for the susceptible population (S). Intuitively, this number decreases over time as individuals contract the disease and leave this group, which accounts for the negative sign. The multiplication of S and I stems from the assumption that the population mixes randomly. It reflects the fact that the likelihood of a disease transmission event depends directly on a susceptible individual coming into contact with an infected one. We divide by the total population size, N, because the term $\frac{I}{N}$ represents the fraction of the population that is currently infected. This fraction gives the realistic probability that a susceptible person will encounter someone who is sick. By dividing by N, we make sure the model assumes people have a normal amount of daily contacts, rather than incorrectly assuming they meet way more people just because they live in a larger population. Finally, the transmission parameter, β, dictates the overall rate at which these infectious contacts occur.

As the number of susceptible individuals decreases, they must transition into the infected group (I). This inflow of newly infected individuals explains the first term on the right-hand side of the second equation (2). At the same time, infected individuals are recovering at a specific rate, γ. This continuous outflow from the infected group accounts for the subtracted second term.

The third equation (3) describes the change in the removed or recovered population (R). It is intuitive that this number—which, as a reminder, encompasses both successful recoveries and disease-induced fatalities—will grow over time. The rate at which this group increases is exactly equal to the rate at which individuals exit the infected group.

3 Numerical Methods

It is important to note that the system of ordinary differential equations governing the SIR model is nonlinear due to the interaction between the susceptible and infected populations. Because of this nonlinearity, these ODEs do not have a simple analytical solution. Therefore, to analyze the dynamics of the epidemic, it is necessary to employ numerical methods to approximate the solutions to these equations.

To demonstrate the implementation of numerical methods in solving the SIR model, this project utilizes a classic benchmark dataset from mathematical epidemiology. In January 1978, an influenza A/H1N1 outbreak spread rapidly through a closed boys' boarding school of 763 students in northern England. Over a strictly recorded 14-day period, the epidemic peaked with nearly 300 students simultaneously infected before naturally subsiding. Because of the perfectly isolated population and precise daily health records, this specific historical event provides an ideal environment for testing and validating foundational compartmental models.

3.1 Euler Method

The Euler method is a fundamental first-order numerical procedure used for approximating the solutions of ordinary differential equations (ODEs) with a given initial value. It calculates the solution curve step-by-step by using the derivative (slope) of the function at a known point to estimate the function's value at a subsequent point.

Given an initial value problem of the form $\frac{dy}{dt} = f(t, y)$ with the initial condition $y(t_0) = y_0$, the Euler method discretizes the time domain into steps of size Δt. The value of the state variable at the next time step, y_{n+1}, is computed using the following explicit formula:

$$y_{n+1} = y_n + \Delta t \cdot f(t_n, y_n) \tag{4}$$

While the Euler method is computationally straightforward and easy to implement, it is a first-order method. Consequently, its accuracy and numerical stability are highly dependent on selecting a sufficiently small time step, Δt, to minimize the local truncation error at each iteration.

The sensitivity analysis presented in Figure 1 illustrates how the fundamental shape of an epidemic is governed by the basic reproduction number ($R_0 = \frac{\beta}{\gamma}$). By systematically adjusting one parameter while holding the other constant, we can isolate their specific effects on the system's dynamics. It is important to note that this behavior is an inherent mathematical property of the SIR model's differential equations, completely independent of the numerical method used to approximate the solution. As shown in the left panel, increasing the transmission rate (β) accelerates the exponential growth phase, resulting

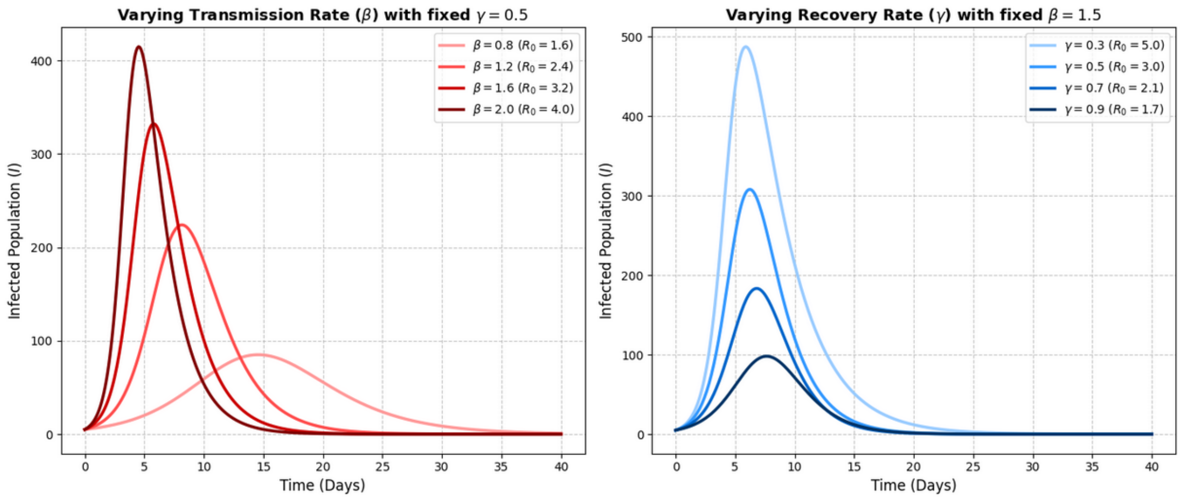

Figure 1: Parameter sensitivity analysis demonstrating the effect of varying transmission (β) and recovery (γ) rates on the infected population. Adjusting these parameters illustrates the mathematical concept of flattening the epidemic curve.

in a higher, earlier peak of active infections. Conversely, the right panel demonstrates that increasing the recovery rate (γ) reduces the infectious window, thereby significantly lowering the peak and mitigating the overall severity of the outbreak.

Figure 2 demonstrates the model's forecasting capability when grounded in real-world historical data. To achieve this, we developed a Python data pipeline utilizing the `scipy.optimize.curve_fit` function to computationally minimize the error between our forward Euler approximation and the actual historical record. By utilizing this optimization algorithm to fit the β and γ parameters strictly to the first week of observed infections (represented by the black dots), the simulation successfully maps the reality of the virus. This deliberate train-test split ensures that the algorithm is not merely curve-fitting the past, but actively forecasting the future. The close alignment between the predicted infected curve (solid red line) and the observed data during the second, "unseen" week mathematically confirms that the model can accurately predict the trajectory and definitive end of an epidemic.

3.2 Runge-Kutta Method (RK4)

The classic fourth-order Runge-Kutta method (often abbreviated as RK4) is a highly accurate and widely used numerical technique for solving ordinary differential equations. While the first-order Euler method relies solely on the derivative at the beginning of a time step, RK4 achieves greater precision by calculating four distinct slopes within each interval and computing their weighted average.

For a given initial value problem $\frac{dy}{dt} = f(t, y)$, the method determines the next value, y_{n+1}, using the following sequence of calculations:

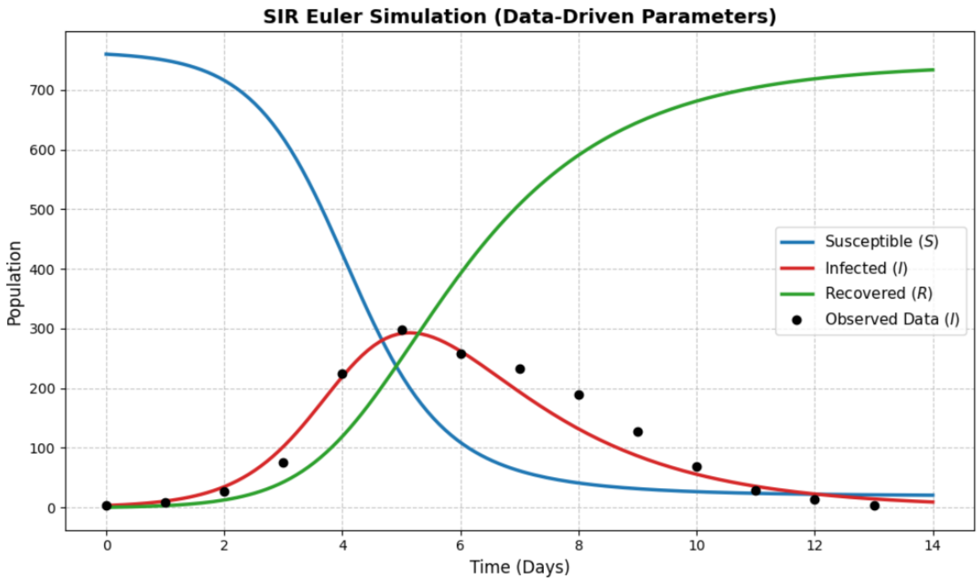

Figure 2: Forward Euler approximation of the SIR model compared against observed data from the 1978 British boarding school influenza outbreak. Model parameters were computationally estimated using only the first 7 days of training data and time stamp dt = 0.1.

$$k_1 = f(t_n, y_n) \tag{5}$$

$$k_2 = f\left(t_n + \frac{\Delta t}{2}, y_n + \frac{\Delta t}{2}k_1\right) \tag{6}$$

$$k_3 = f\left(t_n + \frac{\Delta t}{2}, y_n + \frac{\Delta t}{2}k_2\right) \tag{7}$$

$$k_4 = f(t_n + \Delta t, y_n + \Delta t k_3) \tag{8}$$

Once these four slopes are evaluated, the state at the next time step is computed as:

$$y_{n+1} = y_n + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4) \tag{9}$$

In this weighted average, the midpoint slopes (k_2 and k_3) are given twice the weight of the endpoints (k_1 and k_4). Because RK4 is a fourth-order method, the global truncation error is on the order of $O(\Delta t^4)$. This makes it vastly more accurate and stable than the Euler method, allowing for larger time steps without sacrificing precision.

To demonstrate the efficacy of the RK4 method, the numerical solver was applied to the 1978 boarding school influenza dataset. Figure 3 displays the RK4 approximation using data-driven parameters. The transmission (β) and recovery (γ) rates were computationally optimized using only the first seven days of training data. As shown, the RK4 solver seamlessly captures the epidemic's exponential peak and subsequent decline, proving its reliability in forecasting real-world transmission dynamics.

The theoretical advantage of the RK4 method becomes highly apparent when evaluating numerical stability across larger time steps. As illustrated in Figure 4, both the

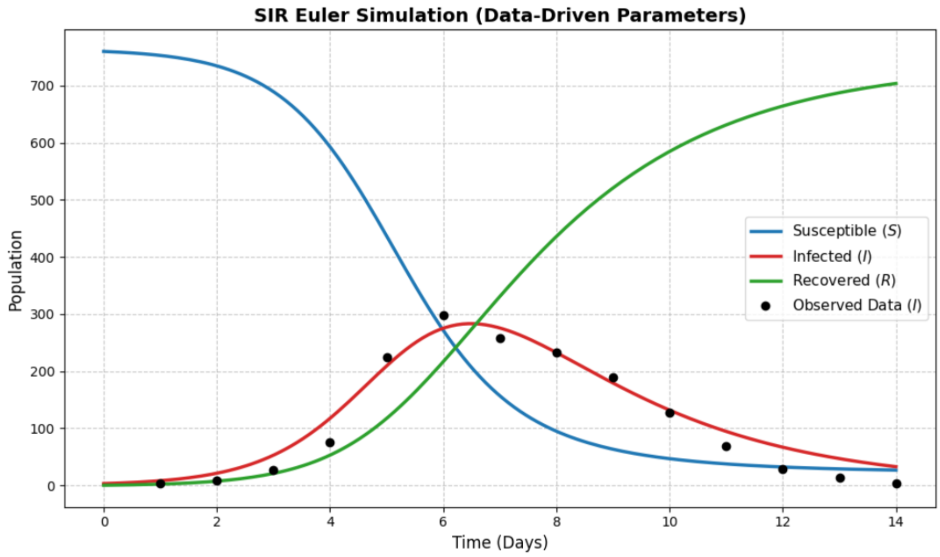

Figure 3: RK4 approximation of the SIR model compared against observed data from the 1978 British boarding school influenza outbreak. The continuous curve accurately forecasts the unseen second week of the epidemic.

Forward Euler and RK4 methods were tested against a high-precision baseline using an intentionally large step size of $\Delta t = 1.5$.

Because the Euler method is restricted to a first-order linear approximation, it evaluates the rate of infection at the start of the step and draws a straight line forward, failing to account for any curvature. Consequently, it drastically overshoots the true mathematical trajectory. Conversely, the RK4 method successfully anticipates the curvature of the epidemic by checking the midpoints of the time interval. Because of this sophisticated averaging, RK4 provides vastly superior numerical stability, perfectly tracking the high-precision baseline despite the massive time step.

To fully evaluate the performance of these numerical solvers beyond visual stability, we must analyze how their errors decrease as we shrink the time step (Δt). Figure 5 displays this relationship side-by-side using both linear and logarithmic scales.

The left panel utilizes a standard linear scale. On this graph, the Forward Euler method forms a straight diagonal line because its error decreases linearly with the time step. In contrast, the RK4 error appears completely flat and indistinguishable from zero for most step sizes. While this linear plot intuitively demonstrates that RK4 is vastly more accurate in absolute terms, it squashes the high-precision data, making it impossible to mathematically measure the algorithm's rate of improvement.

To solve this, the right panel utilizes a log-log scale. In numerical analysis, a solver's error E is proportional to the time step Δt raised to a power p, which represents the theoretical order of the method:

$$E \approx C \cdot \Delta t^p$$

By taking the logarithm of both sides, this power-law relationship is transformed into a linear equation:

$$\log(E) \approx p \cdot \log(\Delta t) + \log(C)$$

On a log-log plot, the slope of the line directly reveals the exponent p. The graph shows the Forward Euler line has a slope of roughly 0.95, confirming its first-order theoretical

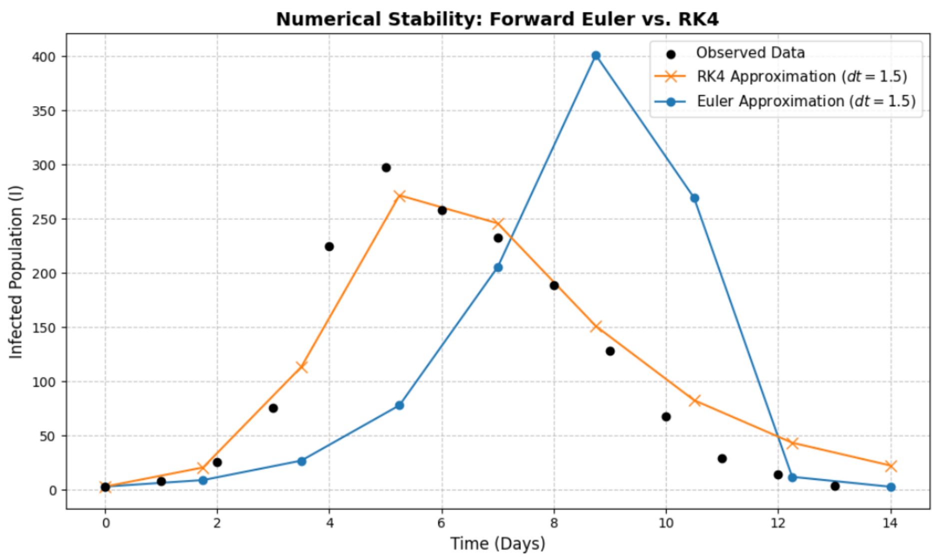

Figure 4: A visual demonstration of numerical stability. When evaluated with a large time step ($\Delta t = 1.5$), the Forward Euler method accumulates significant error and overshoots the peak, whereas the RK4 method remains highly precise.

bound. The RK4 line exhibits a slope of 3.72, clearly demonstrating its fourth-order accuracy. This mathematically proves that halving the time step in RK4 reduces the error by a factor of roughly 16, making it a vastly superior engine for epidemic modeling.

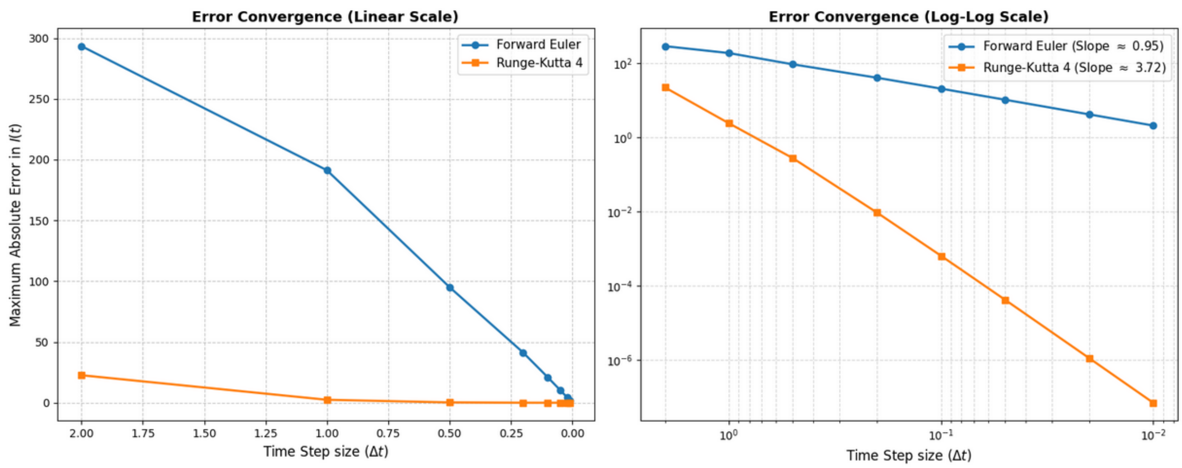

Figure 5: Side-by-side comparison of error convergence. The linear scale (left) highlights the absolute difference in accuracy, while the log-log scale (right) un-squashes the data to reveal the mathematical order of convergence (the slope) for each method.

4 Limitations of SIR model

While numerical methods such as the 4th Order Runge-Kutta (RK4) algorithm provide highly stable solutions for the ordinary differential equations (ODEs) of the SIR model, the underlying mathematical assumptions of the basic framework severely restrict its applicability to complex, long-term epidemiological data.

To explore these limitations, the model was evaluated against real-world data tracking the COVID-19 outbreak in Serbia throughout 2021. This dataset, sourced from the open-access database maintained by *Our World in Data* (OWID), provides the daily number of newly confirmed cases.

In a theoretical SIR model, the infected compartment, $I(t)$, represents the total number of individuals who are currently sick and actively transmitting the virus. However, because public health agencies rarely track the exact duration of individual infections, true "active case" data is largely unavailable. To address this, daily new cases are utilized as a practical proxy. Although the absolute numbers are lower than total active infections, the shape of the incidence curve—its growth rate, peak timing, and eventual decline—closely mirrors the true infection trajectory. This makes it a highly valid metric for testing the model's behavior.

When the basic SIR model is applied to this 360-day dataset—a timeframe characterized by two distinct epidemic waves—its structural limitations become immediately apparent. As visualized in Figure 6, the optimizer completely fails to capture the multi-wave dynamics, collapsing the simulation into an almost horizontal line.

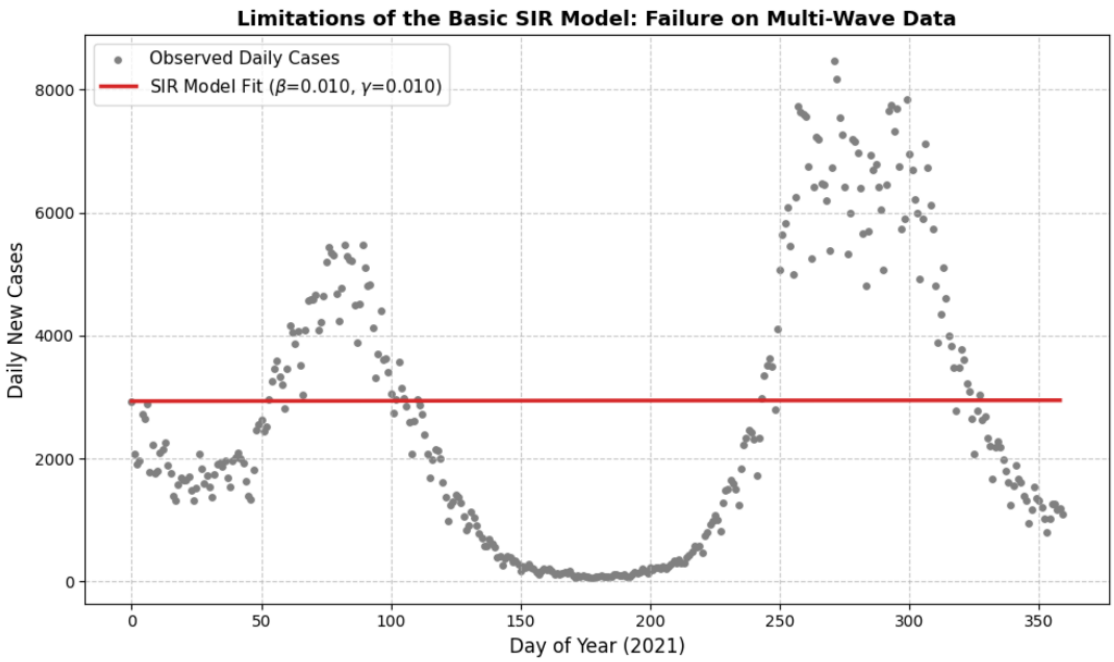

Figure 6: Failure on Multi-Wave Data. The model defaults to a constant average to minimize the global error across multiple peaks.

4.1 Mathematical Constraints of a Closed System

The main limitation of the standard SIR model comes from its equation for infected people:

$$\frac{dI}{dt} = \left(\frac{\beta S}{N} - \gamma\right) I$$

In simple terms, this math means the number of infections can only go up once, peak, and then go down. As long as there are many healthy people (the susceptible population, S) available to catch the virus, the infection spreads very quickly. However, once enough people get sick and recover, the virus runs out of new targets to infect. This turning point is known as herd immunity.

After reaching this point, the number of active infections drops steadily until the outbreak disappears completely. Because of how this equation is built, the basic SIR model is only capable of drawing one single epidemic wave and cannot bounce back up.

4.2 Optimization Breakdown in Non-Convex Landscapes

When a computational optimization algorithm, such as the Levenberg-Marquardt method used in `scipy.optimize.curve_fit`, attempts to minimize the Sum of Squared Errors (SSE) over a multi-wave dataset, it faces a structural impossibility. Fitting the parameters to perfectly capture the first wave would result in an extreme error penalty during the second wave. To minimize the global error across both peaks and the intervening trough, the optimizer defaults to a degenerate solution: it drives both the transmission rate (β) and the recovery rate (γ) to their absolute lower bounds, effectively freezing the rate of change and producing a flatline average.

4.3 Epidemiological Implications

The failure of the model over the full year highlights the violation of its core assumptions in real-world scenarios. The basic SIR model assumes a constant transmission rate (β) and permanent immunity upon recovery. However, macroscopic epidemiological events are heavily influenced by dynamic human behavior (e.g., lockdowns), the emergence of more contagious virus variants, and waning immunity. To accurately model multi-wave outbreaks, the mathematical framework must be extended.

4.4 Successful Modeling of Localized Outbreaks

Despite its failure over a full-year timeframe, the model is highly effective when applied to specific, localized parts of the dataset. If the evaluation period is restricted to a single epidemic wave, the RK4 numerical solver successfully captures the trajectory of the outbreak.

As shown in Figure 8, isolating the dataset to a single surge (approximately between days 30 and 130) allows the theoretical curve to closely align with the observed real-world daily cases. This localized success is entirely logical and aligns perfectly with the mathematical constraints previously discussed. Because the underlying equations are restricted to generating exactly one peak, the model can successfully fit the data as long as the observation window is limited to a single wave where the transmission rate (β) remains temporarily constant.

To rigorously assess the performance of the RK4 solver on this localized epidemic wave, the fit is quantified using a combination of standard statistical error metrics and domain-specific epidemiological checks based on the data presented in Figure 8:

- **Root Mean Squared Error (RMSE):** Measures the standard deviation of the residuals. Because RMSE squares the errors before averaging them, it heavily penalizes large misses, making it ideal for highlighting how a model fits the height of an epidemic peak. **RMSE: 504.86**

- **Mean Absolute Error (MAE):** Measures the average absolute distance between the observed data and the predicted curve. MAE treats all errors equally, making it less sensitive to isolated reporting outliers and providing a clean view of the model's average day-to-day deviation. **MAE: 390.12**

- **Peak Timing and Magnitude Error:** Evaluates the model's practical forecasting ability. Peak Timing Error measures the difference in days between the real and predicted peaks, while Peak Magnitude Error measures the difference in the maximum height of the wave. **Peak Timing Error: 5 days | Peak Magnitude Err: 850 cases (15.5%)**

- **Coefficient of Determination (R^2):** Represents the proportion of variance in the dependent variable that is mathematically predictable from the model. R^2**: 0.8459**

- **Akaike and Bayesian Information Criteria (AIC/BIC):** Evaluate goodness-of-fit while penalizing model complexity (number of parameters). These ensure the underlying mathematics are genuinely improved rather than simply overfitted to data noise. **AIC: 1236.41 | BIC: 1241.60**

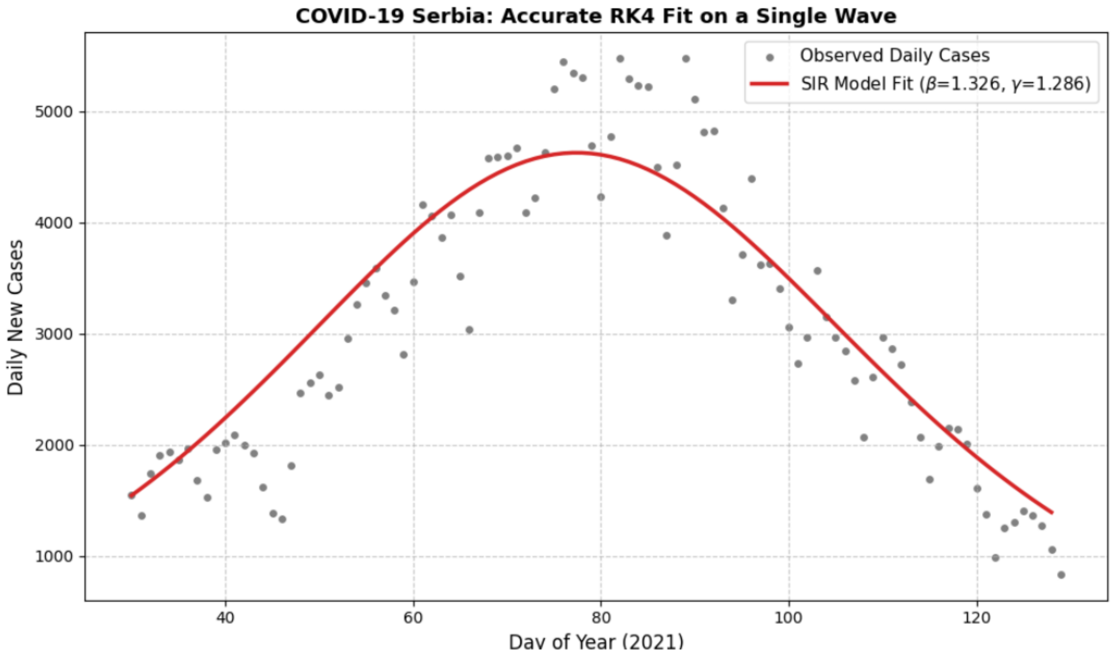

Figure 7: COVID-19 Serbia: Accurate RK4 Fit on a Single Wave. The basic SIR model successfully captures the outbreak dynamics when the data is restricted to a single peak.

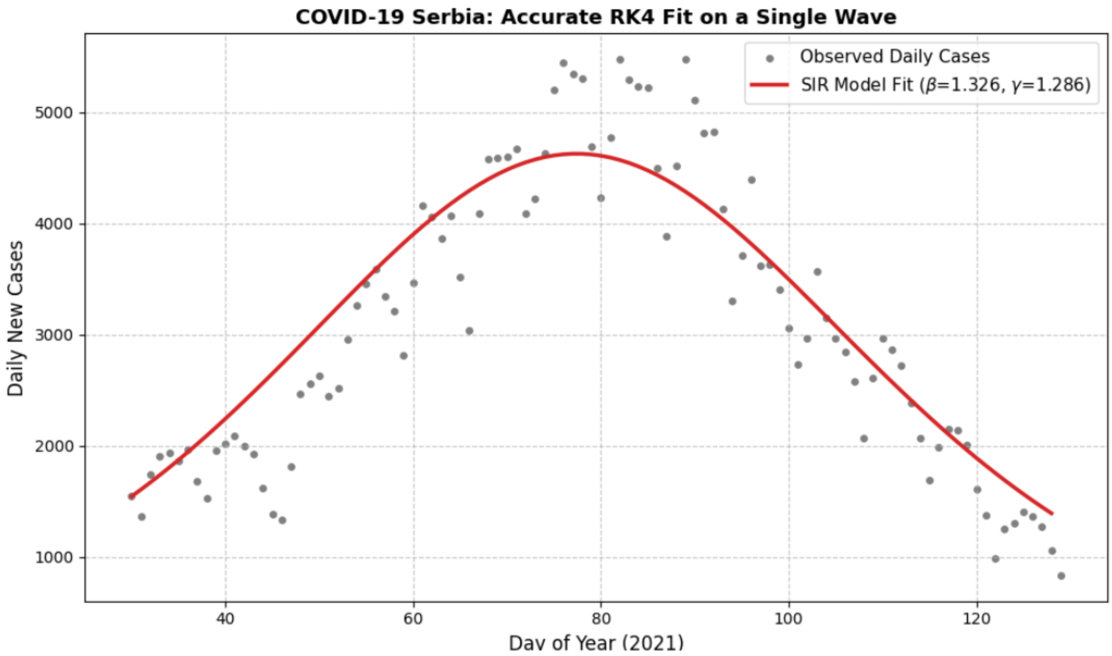

Figure 8: COVID-19 Serbia: Accurate RK4 Fit on a Single Wave. The basic SIR model successfully captures the outbreak dynamics when the data is restricted to a single peak.

5 Conclusion

This project evaluated the application of numerical methods to the SIR compartmental model, concluding that the fourth-order Runge-Kutta (RK4) algorithm provides vastly superior stability and precision compared to the Forward Euler method for solving epidemiological ODEs. When combined with computational parameter optimization, the RK4 solver accurately maps isolated, single-wave outbreaks, such as the 1978 boarding school influenza and short-term COVID-19 surges. However, the basic SIR framework inherently breaks down on long-term, multi-wave datasets. Because its closed-system equations permit only a single infection peak before reaching herd immunity, modeling prolonged pandemics requires extending the mathematical framework to account for dynamic human interventions, viral mutations, and waning immunity.

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